



Quantum Computing for Machine Learning

Application to Network Traffic Classification
and Zero-Day Attack Detection

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Classical Machine Learning Already Works

- But classical methods have a cost:



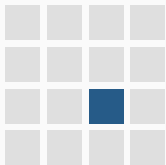
So what is the open challenge?

Finding a setting where a *different* kind of computer offers something classical models cannot.

The Promise of Quantum Machine Learning

Classical

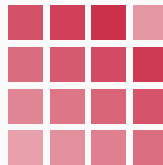
one state at a time



1 active state

Quantum

many states at once (superposition)



many states explored in parallel

With **superposition** and **entanglement**, a quantum circuit explores **many states in parallel**.

The premise — a small quantum model could capture patterns a classical model of the same size cannot.

The Open Question

On a **realistic data-science problem**,
can a parameterized quantum circuit do something
that a strong classical model *cannot*?

Existing answers

synthetic data
or theoretical proofs

VS

This TER

real IoT network traffic
empirical comparison

Qubit & Superposition

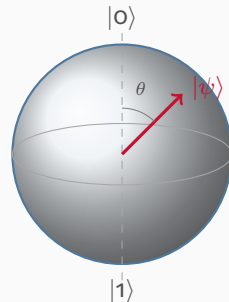
- A classical bit is 0 or 1.
- A qubit can be in a **superposition**:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

- With n qubits, the state space has size 2^n .

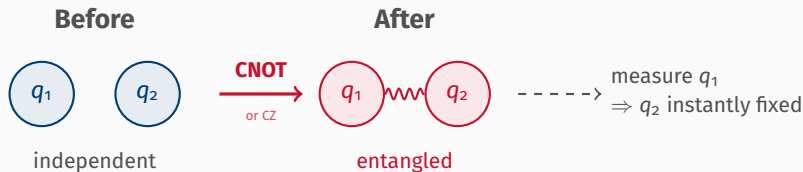
A classical register sits in *one* state.

A quantum register can hold a *combination* of all 2^n states at once.



Bloch sphere — every pure qubit state is a point on it.

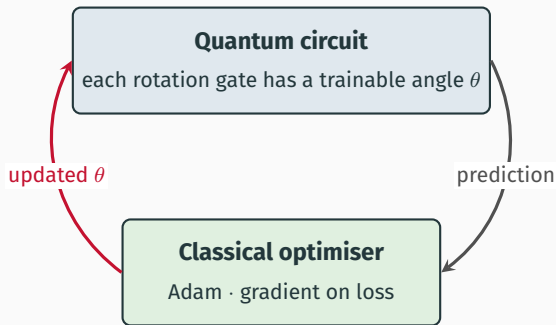
Entanglement



Why it matters for ML

Entanglement gives the circuit *joint expressive power*: it can encode patterns *between* features that single-qubit gates cannot.

Trainable Rotations



This gives us three things:

1. **Angle encoding**
data $\rightarrow$ angles
2. **Short circuit**
real hardware is noisy
3. **Hybrid loop**
quantum forward + classical update

θ is the **quantum analog of a neural-network weight**.

What This TER Tests

Subject

implement quantum ML models in PennyLane,
apply them to a network dataset

Dataset

ToN_IoT
intrusion detection on IoT traffic

Baselines

classical (SVM, RF, NNs)
+ an unsupervised autoencoder

Quantum architectures

data re-uploading
QCNN

Positioning

A controlled *data-science* comparison — not a quantum-hardware demonstration.

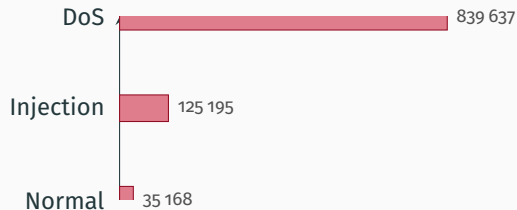
Can a quantum machine learning model detect
injection attacks
when it was trained only on
normal and **DoS** traffic?

This is closer to *anomaly detection* than to standard classification.
*supervised **cannot classify** the unseen — but it may still **flag it as “not normal”**.*

The Dataset: ToN_IoT

Class distribution

~1 million flows — heavily unbalanced

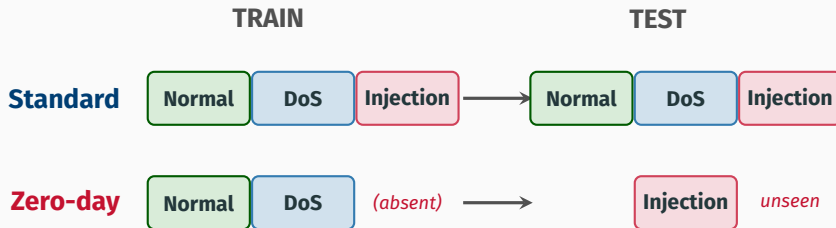


Each row = one network flow: src/dst IP and port, protocol, byte and packet counts, duration, DNS fields.

Preprocessing

- Balance by undersampling: 35,168 per class.
- ANOVA feature selection: 8 features.
- **Train/test split before scaling:** no leakage.

Two Evaluation Settings



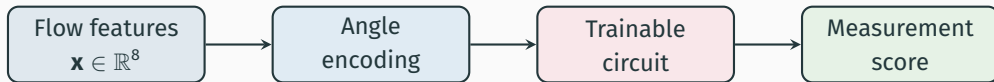
Why this matters

The zero-day setting forces the model to face a class it has *never* seen during training. This is the heart of the TER.

Compared Models

	Classical	Quantum
Supervised	SVM (RBF) Random Forest NN-Small NN-Medium	QCNN (8 qubits) Re-uploading 1q Re-uploading 4q
Unsupervised	Autoencoder <i>trained on normal only</i> <i>95th-percentile threshold</i>	— not in scope

Quantum ML Pipeline



8 features

feature → **rotation angle**

param. rotations
+ entangling gates

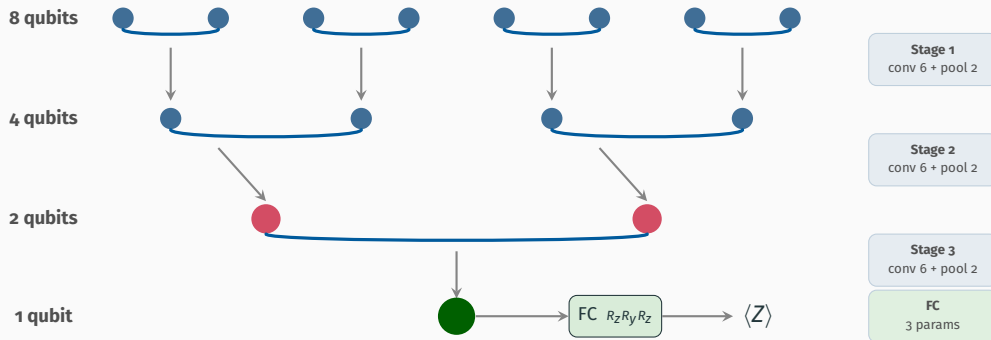
measure output

hybrid loop circuit → prediction, classical Adam → update

PennyLane lightning.qubit, adjoint differentiation, 3× one-vs-all classifiers, no noise.

Architecture (1/2) – QCNN

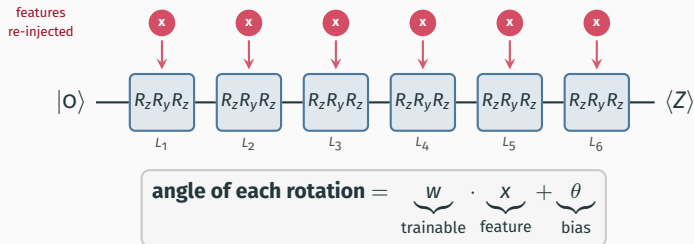
conv = neighbours interact (shared 6-param gate) • **pool** = merge each pair, keep one qubit (CRZ + CRY)



Total: 27 params per binary classifier = 3×6 conv + 3×2 pool + 3 FC

Architecture (2/2) – Data Re-uploading

Key idea: features are re-injected into the qubit at every layer



- **1-qubit variant:** 1 qubit, 6 layers.
- **4-qubit variant:** 4 qubits, 3 layers, CZ entanglement.
- 108 / 216 params per binary classifier.
- 3 independent one-vs-all classifiers (same as QCNN).

Methodology – Three Commitments

1

Data-science controls

Split first.
Fit preprocessing
on train only.

2

Same feature pipeline

Identical features
for every model.
Sample budgets
stated honestly.

3

Honest reporting

10 seeds, full $\pm$ SD,
and failure modes.

Classical models dominate. Random Forest reaches 99%; quantum models stay between 68 and 75%.

With 8 well-selected features, the standard 3-class task is already easy for classical methods.

This is the *expected* result on closed-set tabular data.

Standard Classification – Numbers

Model	Accuracy	F1	MCC
Random Forest	0.990	0.990	0.985
SVM (RBF)	0.970	0.970	0.955
NN-Medium	0.960	0.960	0.940
NN-Small	0.925	0.925	0.888
Re-uploading 1q	0.752 ± 0.039	0.720 ± 0.051	0.662 ± 0.045
Re-uploading 4q	0.692 ± 0.011	0.654 ± 0.010	0.586 ± 0.016
QCNN	0.679 ± 0.025	0.636 ± 0.021	0.559 ± 0.043

Same ranking across accuracy, F1, and MCC. No evidence of quantum advantage here.

Every supervised model fails below 3%.

Two methods reach ~70%: the autoencoder and the QCNN.

Zero-day is no longer just classical vs quantum.

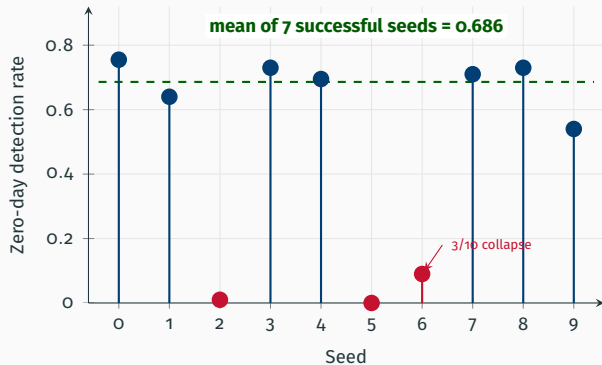
It is *supervised classification vs anomaly detection vs supervised transfer*.

Zero-Day Detection – Numbers

Model	Regime	Detection rate	F1
Autoencoder	unsupervised	0.705 ± 0.144	0.819 ± 0.100
QCNN, 7/10 convergent seeds	supervised transfer	0.686 ± 0.068	0.812 ± 0.051
QCNN, all 10 seeds	supervised transfer	0.490 ± 0.305	0.587 ± 0.349
Re-uploading 1q	supervised	0.092 ± 0.036	0.167 ± 0.058
Re-uploading 4q	supervised	0.075 ± 0.023	0.138 ± 0.039
Best supervised classical (NN-Small)	supervised	0.028	0.054

Test set is injection-only: detection rate = accuracy = injection recall. “Supervised” = trained on normal-vs-DoS labels.

QCNN Seed Sensitivity



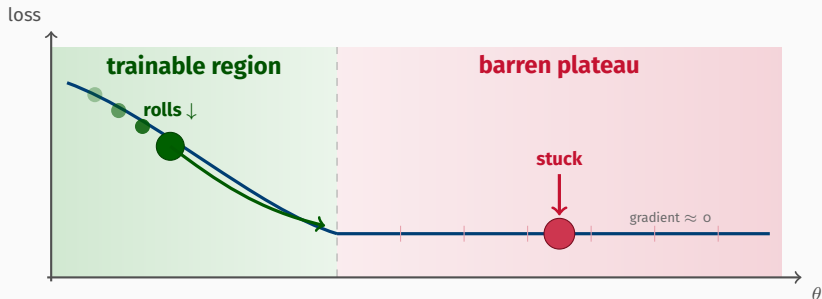
Pattern

- 7 seeds converge: 54–76%.
- 3 seeds collapse: $\leq 9\%$.

Observed rates

- QCNN collapse: 30% (3/10).
- Autoencoder failure: 10% (1/10).

Cause – Barren Plateau



The honest take

I report all 10 seeds, not only the best run.

A single seed could give a misleading conclusion in either direction.

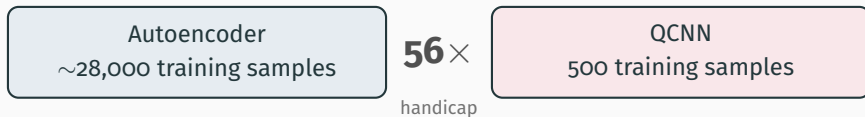
- The loss landscape becomes flat — *no slope, no direction*.
- Training collapses from a bad start — a *known* phenomenon.

Interpretation – Autoencoder vs QCNN

	Autoencoder	QCNN
	<i>practical baseline</i>	<i>scientific observation</i>
Training data	normal only	supervised normal-vs-DoS
Mechanism	high reconstruction error	quantum feature transfer
Detection rate	70.5%	68.6% (7/10 seeds)

The QCNN result is scientifically interesting, but the autoencoder remains the more *reliable* practical detector today.

An Asymmetry to Keep in Mind



- **Despite** this handicap, the QCNN reaches comparable detection.
- **Therefore** — not a strict head-to-head.
- **Future** — closing the gap requires real quantum hardware.

QCNN training size is bounded by simulation cost, not by the algorithm itself.

What I Do Not Claim

No quantum advantage on the standard task

No practical win autoencoder still ahead

No hardware claim noiseless simulation only

Limited seeds 10 reveal instability; more would sharpen

Methodological

what to improve

More seeds — 30 — 50

Barren-plateau mitigation

Realistic noise simulation

Scaling & scope

what to expand

Larger circuits, more qubits

Real quantum hardware

Broader attack types

backdoor, ransomware, scanning

In all cases — same protocol, multi-seed evaluation, comparison against the autoencoder.

The promise of quantum ML **survives in part.**

Findings

what we observed

Standard task: **classical methods win**

Zero-day: classical supervised methods **fail**

Takeaway

what it means

Autoencoder: **practical** baseline

QCNN: *fragile but real* transfer

Quantum ML is **not ready to replace classical methods today** — but it deserves careful study.

Key References

Architecture anchors

- **Pérez-Salinas et al. (2020)** – *Data re-uploading for a universal quantum classifier*. Quantum.
- **Hur, Kim & Park (2022)** – *Quantum convolutional neural network for classical data classification*. Quantum Machine Intelligence.

Theoretical anchor

- **McClean et al. (2018)** – *Barren plateaus in quantum neural network training landscapes*. Nature Communications.

Other: Moustafa (2021) – ToN_IoT dataset; Benedetti et al. (2019); Cerezo et al. (2021); Preskill (2018) – PQC and NISQ context.

Thank you

for your attention.